Hybrid UAV Waypoint Tracking Simulation, Results Anlysis and Optimization

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Abstract

This document focuses on the simulation results, analysis and corresponding optimizations of a pid based simple "Hybrid UAV Waypoint Tracking" (square climbing) task simulated by Panerati et al. [2021].

1 Mathematical Model

1.1 State Space (in 3D space)

For *Continuous State*:

- Position: $p \in \mathbb{R}^3$
- Velocity: $v \in \mathbb{R}^3$

For *Discrete State*:

- Index of Waypoint (Tracking Target): $q \in \mathcal{Q} = \{0, 1, 2, \ldots, N\}$
  - Coordinates $W = \{W_0, W_1, \ldots, W_N\}$, where $W_q \in \mathbb{R}^3$ is the coordinate of idx q

Thus, the complete **hybrid state vector $\boldsymbol{x}$** of the system is defined as:

$$x = \begin{bmatrix} p \\ v \\ q \end{bmatrix} \in \mathbb{R}^6 \times \mathcal{Q} \tag{1}$$

1.2 Flow Set C

The flow set determines the conditions under which the system undergoes continuous-time physical evolution:

- As long as the Euclidean distance between the UAV and the current target waypoint W_q is greater than or equal to the tolerance radius ϵ
- or if the system has reached the final waypoint ($q = N$), it maintains continuous flight:

$$C = \left\{ x \in \mathbb{R}^6 \times \mathcal{Q} \mid \|p - W_q\| \geq \epsilon \vee q = N \right\} \tag{2}$$

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1.3 Flow Map f

Within the flow set, the evolution of the system follows the **differential equation $\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x})$**. During this phase:

- For continuous state:
  - $\dot{p} = v$
  - $\dot{v} = \mathcal{F}_{dyn}(p, v, u_{pid}(p, v, W_q))$
- For discrete state q: $\dot{q} = 0$ since q is invariant in flow set
- Physical state is governed jointly by the underlying *rigid body dynamics* and the *closed-loop PID controller*:

$$\dot{x} = f(x) = \begin{bmatrix} v \\ \mathcal{F}_{dyn}(p, v, u_{pid}(p, v, W_q)) \\ 0 \end{bmatrix}, \quad x \in C \tag{3}$$

Note:

- u_{pid} is the *control command* with W_q acting as the absolute target
- $\mathcal{F}_{dyn}$ represents the *rigid body dynamics integration* computed by the underlying *6-DOF physics engine*

1.4 Jump Set (D)

The jump set defines the strict boundary where discrete events (*transients*) are triggered:

- When the UAV enters the ϵ-neighborhood of the target waypoint
- and the current waypoint is not the final destination

The state machine triggers a jump:

$$D = \left\{ x \in \mathbb{R}^6 \times \mathcal{Q} \mid \|p - W_q\| < \epsilon \wedge q < N \right\} \tag{4}$$

1.5 Jump Map (g) (*i.e. Transition*)

When the state $x \in D$, the system undergoes a discrete jump $x^+ = g(x)$.

- At this instant, the physical position and velocity remain continuous (**no spatial mutation occurs**)
- but the target waypoint index q advances to the next one:

$$x^+ = g(x) = \begin{bmatrix} p \\ v \\ q+1 \end{bmatrix}, \quad x \in D \tag{5}$$

1.6 Overall Hybrid Automaton Equations

$$\mathcal{H} := \begin{cases} \dot{x} = f(x), & x \in C \\ x^+ = g(x), & x \in D \end{cases} \tag{6}$$

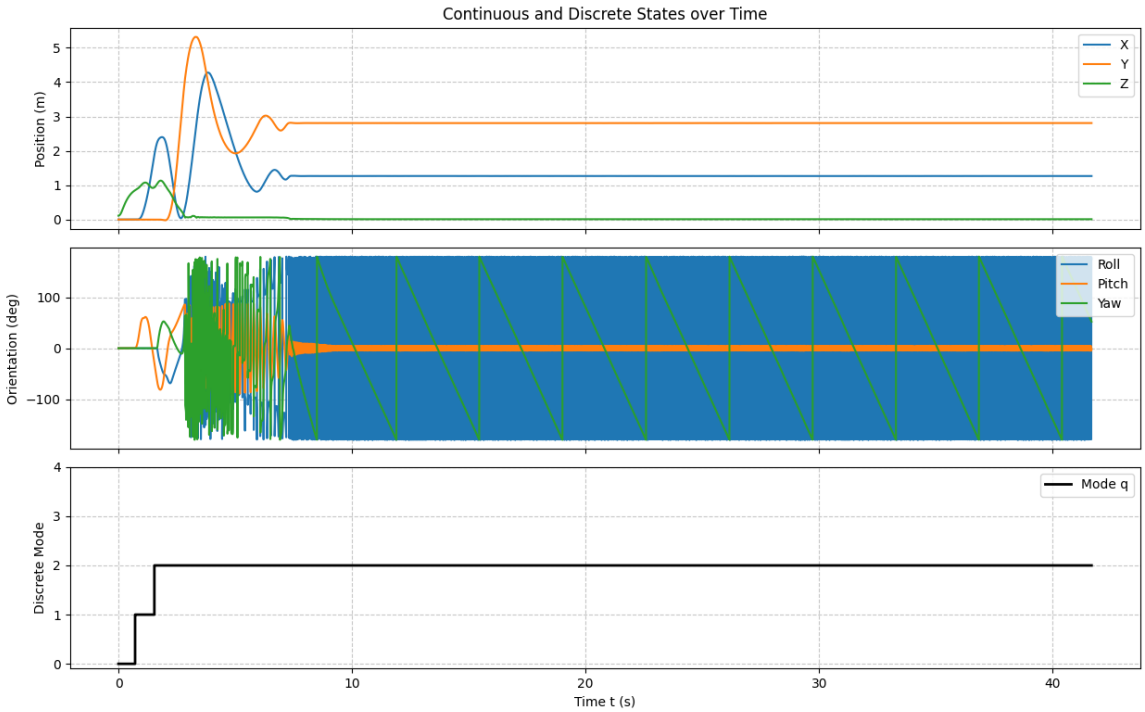

Figure 1: Initial Exp: State Plot

2 Simulation Results

2.1 Initial Exp

State & Trajectory Plot The initial experiment's state plot and trajectory plot see fig. 1 and fig. 2.

2.2 Simulation Results Analysis

Conclusion: The breakdown of local stability in **underactuated systems** under large step inputs.

Overview Results Analysis: As observed in the state and trajectory plots, the UAV successfully tracks the initial waypoints, indicated by the discrete mode transitions ($q = 0 \to 1 \to 2$). However, upon switching to mode $q = 2$ (targeting [2.0, 2.0, 1.5]), the large spatial distance introduces a severe step-error to the closed-loop controller. This forces an aggressive attitude adjustment, evident from the violent high-frequency oscillations in the Orientation plot (Roll/Pitch/Yaw) after $t \approx 2s$. The extreme tilt angles cause the vertical thrust component to drop below gravity ($T \cos\theta < mg$), leading to an irreversible altitude loss ($Z \to 0$, as seen in the Position plot). The UAV hits the ground and becomes locked in an unrecoverable physical state, failing to ever enter the ϵ-neighborhood (Jump Set D) of the subsequent waypoint. Consequently, the discrete state q permanently stalls at 2.

Potential Dynamical Properties to Study:

- **Stability (Region of Attraction):** The continuous flow map $f(x)$ utilizing standard PID control is only *locally asymptotically stable* for **the 6-DOF underactuated system**. It exhibits severe instability under large step inputs due to the coupling between attitude and translational dynamics.
- **Attractivity (Reachability):** The target jump set D_{q+1} is not attractive from the post-jump initial state $x(t_0)$ under the current flow map. The continuous trajectory diverges to the ground singularity before intersecting D.

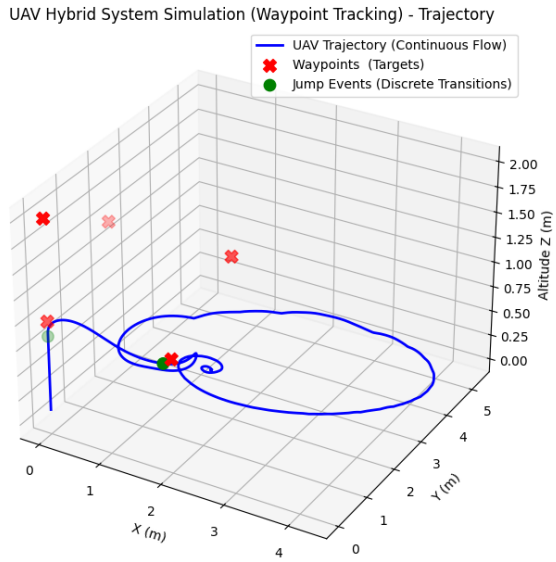

Figure 2: Initial Exp: Trajectory Plot

- **Robustness:** The current system architecture lacks robustness against actuator saturation (motor RPM limits) and unmodeled ground-effect disturbances when subjected to extreme attitude commands.

Outline of a Possible Analysis Approach:

- **Step 1: Lyapunov-based Stability Analysis:** Construct a Control Lyapunov Function (CLF), denoted as $V(x)$, for the closed-loop continuous dynamics to estimate the **Maximum Region of Attraction (ROA)** regarding the allowable positional tracking error.
- **Step 2: Forward Reachability Verification:** Mathematically verify whether the initial state immediately following a discrete jump, $x^+ = g(x)$, safely resides within the established ROA of the subsequent flow map.
- **Step 3: Trajectory Redesign (Controller Synthesis):** Introduce a sufficiently smooth reference trajectory planner (e.g., a minimum-snap or cubic polynomial interpolator) into $f(x)$. This ensures the instantaneous tracking error remains strictly bounded within the ROA, guaranteeing forward invariance and successful intersection with all subsequent jump sets.

2.3 Optimization Exp

Optimizations

- **Virtual Target Initialization:** Introduced a *virtual_target_pos* state variable initialized exactly at the drone's spawn location:
  - This ensures the **initial control error is zero**,
  - **preventing any violent takeoff maneuvers**.

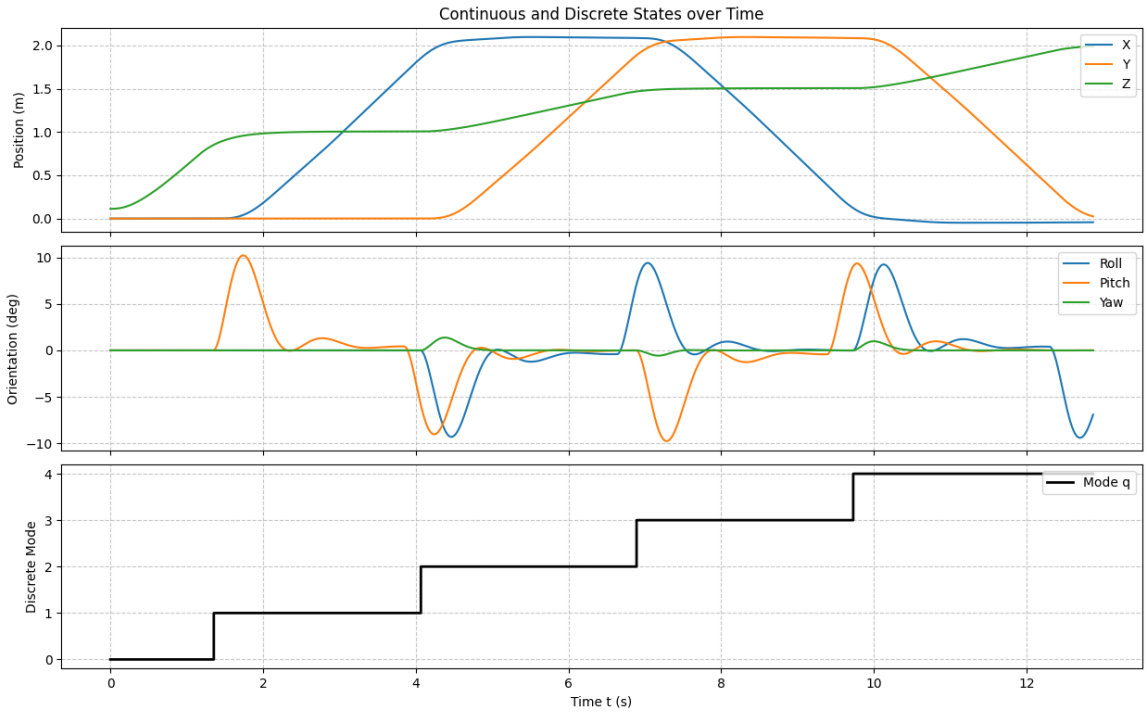

Figure 3: Optimization Exp: State Plot

- **Kinematic Trajectory Generator (Linear Interpolation):** Added a **smooth interpolation mechanism** *inside* the continuous Flow Map $f(x)$ loop. Instead of instantly switching to the next distant waypoint, the virtual target moves towards the absolute waypoint at a **constrained, constant velocity** (MAX_SPEED = 0.8 m/s). -> Enhance Stability
- **Elimination of Step-Inputs for the Controller:**
  - Modify the PID controller to track this continuously moving *virtual_target_pos* rather than the absolute *WAYPOINTS[q]*.
  - By breaking down the large spatial gap into millimeter-level, time-stepped increments, we completely eliminated the massive step-errors that previously caused extreme attitude (pitch/roll/yaw) oscillations and the local stability failure of the underactuated system.

State & Trajectory Plot & GUI Display The optimization experiment's state plot, trajectory plot and gui display see fig. 3, fig. 4 and fig. 5.

References

J. Panerati, H. Zheng, S. Zhou, J. Xu, A. Prorok, and A. P. Schoellig. Learning to fly—a gym environment with pybullet physics for reinforcement learning of multi-agent quadcopter control. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 7512–7519, 2021. doi: 10.1109/IROS51168.2021.9635857.

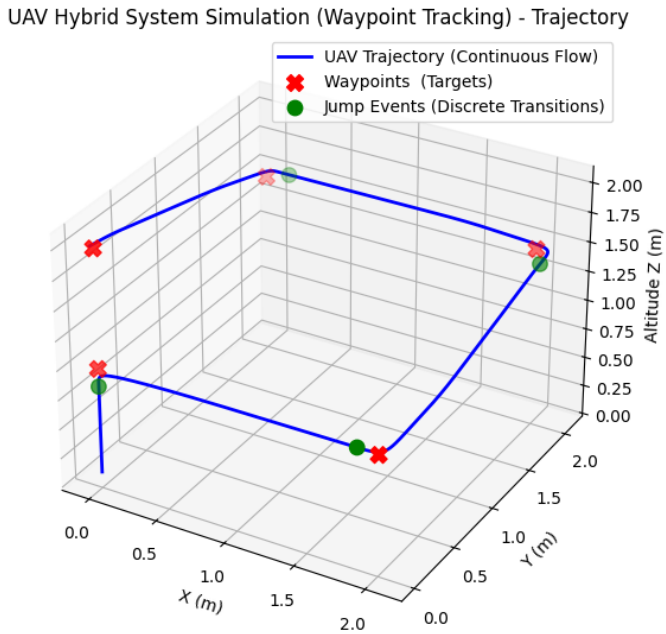

Figure 4: Optimization Exp: Trajectory Plot

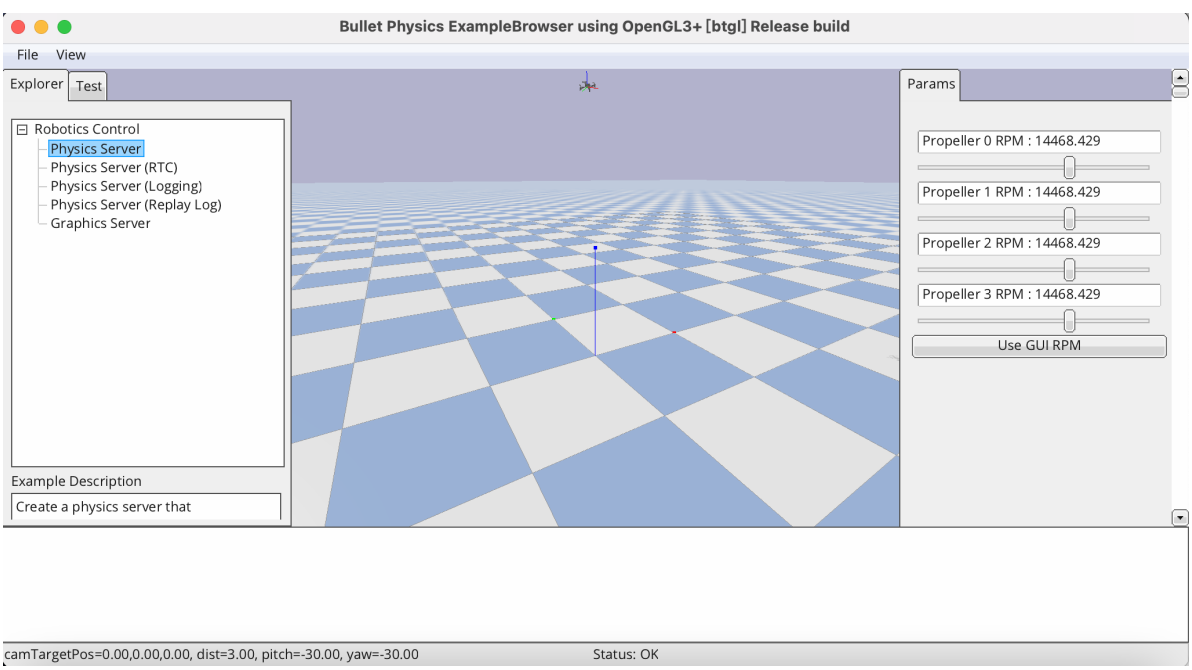

Figure 5: Optimization Exp: Simulation GUI Display